

Lyu Zhipeng

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EDUCATION BACKGROUND

Guangdong University of Petrochemical Technology (GDUPT) 09/2021-06/2025

Bachelor of Engineering

Major: Data Science and Big Data Technologies

Lingnan University 09/2025-06/2026

Master of Science

Major: Data Science

INTERNSHIP EXPERIENCE

Shenzhen Guangda Tongchuang New Material Co., Ltd. 08/2024-09/2024

IT Intern

- Responsible for desktop and software technical support, maintaining computer hardware, software, and network connections to ensure stable operation of office systems.
- Assist in server and database management, participate in data backup and recovery, complete company-wide office computer system updates, check compatibility and execute updates, test application availability, and record issues and solutions
- Participate in internal tool development, write code, fix bugs, and support business process digitization

Zhejiang Haikan Technology Group Co., Ltd 07/2024-08/2024

Big Data Intern

- Responsible for collecting, cleaning, and validating business data, we automate the handling of missing and outlier values using Python scripts, boosting data availability to over 95% and laying a solid foundation for subsequent analysis.
- Utilize Python (Pandas/NumPy) to perform multidimensional analysis of user behavior and business data, extracting key metrics and trends, and generating analytical reports to provide data-driven support for product optimization.
- Create interactive data dashboards with Tableau to visualize analysis results and help teams quickly understand business dynamics. Participate in the entire data project process, collaborate with product and development teams, and drive the implementation of analysis conclusions.

Gaozhou Fengsheng Food Co., Ltd. 07/2023-08/2023

Intern Assistant Engineer

- Assist R&D personnel in the research lab to analyze and process display parameters and accuracy, organize the analysis results, and provide data support for product development.
- Responsible for database creation and data entry, standardizing data formats to ensure accuracy and traceability.
- Collaborate with the team to conduct literature reviews and market research, gather and organize industry data, and support R&D decision-making.

PROJECT EXPERIENCE

Blockchain Electronic Voting System (Smart Contract Development)

11/2025-01/2026

Project Overview: This decentralized voting system built on the Ethereum blockchain utilizes smart contracts to automate user management and the entire voting process, ensuring transparent, traceable data and tamper-proof results. It is ideal for community governance, internal elections, and similar scenarios.

Core Responsibilities:

- Lead the end-to-end smart contract development process, including technology selection, system architecture design, contract development, and testing deployment, while bridging functional implementation with field testing.
- The system adopts a three-tier architecture (front-end interaction layer, smart contract layer, and blockchain layer) to achieve front-end-back-end separation and decentralized trust mechanisms.
- Complete requirements analysis, contract design, code implementation, and testing deployment to provide a stable and verifiable foundation for subsequent system testing.

Technical point:

- Develop smart contracts with Solidity 0.8.0, featuring built-in overflow checks for enhanced security, and utilize the Hardhat framework for efficient compilation, testing, and deployment.
- The front-end utilizes Vue.js and Ethers.js to interact with the MetaMask wallet, delivering a user-friendly DApp interface.
- Design data structures and permission modifiers such as User and VoteActivity to implement core functionalities including user registration, vote execution, and result statistics.

Simulation System of Charging Station Layout in Hong Kong

02/2026-04/2026

Project Overview: This simulation system for charging station deployment in Hong Kong integrates road network data with simulation algorithms, providing data support and visualized decision-making tools for optimizing urban charging station locations and assessing coverage areas.

Core Responsibilities:

- Participated in backend development, assisted in building a graph structure based on OpenStreetMap road network data, implemented the shortest path planning function, and completed testing and validation.
- Responsible for data preprocessing, cleaning CSV node and edge data, and ensuring the format meets backend loading requirements.
- Develop basic API interfaces to support front-end data display; assist in debugging map path rendering issues and fix display errors.
- Participate in the discussion of the location selection optimization algorithm and assist in the implementation of the coverage calculation module.

Technical point:

- The backend service is built using Python and Flask, with the Dijkstra shortest path algorithm implemented through NetworkX.
- Use Pandas to clean and format CSV data to ensure quality.
- Develop RESTful API interfaces (e.g., retrieving charging station lists and electric vehicle locations) for the Leaflet.js front-end map component to use.
- Implement map visualization with Leaflet.js, debug path point rendering logic, and optimize user interaction experience

PERSONAL SKILLS

IT : SQL,Python, C++, KNIME, Hadoop, R, solidity,Tableau

Algorithms: Sorting, Search, Linear Regression

Languages: Chinese (Mandarin-Native), Fluent in English (IELS:6.5).

Self-assessment: Effective communicator with strong adaptability, vitality and responsibility